

INDEPENDENT PEER-REVIEW PACKET

EC-005

Canonical A2 Fold-Gap Identity

Cold Review Packet 1 of 20

Dylan M. Roy Mathematical Research Program

Prepared for independent validation - 27 July 2026

This packet is self-contained for mathematical review. Drive access is supplemental, not required to understand the claim and proof.

READ FIRST

You are being asked to review one narrow terminal object, not the wider q0/P0.2/Theorem B research program. The declared terminal label is an internal research-state classification. Your task is to decide whether the exact statement and its proof/evidence are correct within the stated hypotheses and exclusions.

1. Pass A - Independent reconstruction. Read the exact statement, notation, and hypotheses. Attempt your own derivation or falsification before reading the embedded historical review record.
2. Pass B - Source audit. Compare your reconstruction with the complete controlling source text embedded in Part II. Check every dependency, sign, factor, quantifier, scope boundary, and computational claim.
3. Pass C - Hostile review. Run the object-specific focal checks and invent at least one additional mutation or counterexample attempt.
4. Issue one fail-closed verdict: APPROVE; APPROVE WITH CLARIFICATIONS; AMEND REQUIRED; REJECT; or CANNOT VERIFY.

Do not transfer approval to excluded downstream claims. Do not count access to this packet, the author, or another reviewer as independent mathematical evidence.

Part I - Exact Review Object

Packet	1 of 20
Object ID	EC-005
Object class	Exact algebraic identity
Exact statement	For $f_{\lambda}(x) = -x^3/3 + \lambda x$ with $\lambda > 0$, the critical-point separation $r = 2\sqrt{\lambda}$ and positive critical-value gap ℓ satisfy exactly $\ell = r^3/6$, under the canonical A2 coordinate and normalization.
Internal terminal label	CLOSED - EXACT CANONICAL NORMAL-FORM IDENTITY
Terminal date	2026-07-21
Recommended expertise	Singularity theory / calculus / asymptotic normal forms

1. Significance

This object was separated from larger theorem claims so that its correctness can be assessed without accepting unrelated Gaussian-field, persistence, adjacency, or asymptotic assertions. A positive review supports only the exact statement above and the dependency uses explicitly listed below.

2. Evidence class and prior review

Evidence	Exact symbolic derivation; independent recomputation; hostile scope audit distinguishing the canonical coefficient from the physical-field κ mark.
Prior review record	AO48 and Anthropic/CL independently recomputed the identity; terminal condition was discharged before closure.
Binding exclusions	No universal physical-field coefficient; no lifetime theorem; no adjacency, Palm, or continuum claim.

3. Dependency map and use restrictions

- Inputs: only definitions, hypotheses, and subordinate identities explicitly reproduced in the controlling source text.
- Output: only the exact statement listed in the review-object table.
- Permitted downstream use: as a cited lemma/interface under identical object definitions and hypotheses.
- Forbidden transfer: no approval of a parent theorem, numerical constant, external release, or canonical promotion unless separately reviewed.

- Object-specific exclusions: No universal physical-field coefficient; no lifetime theorem; no adjacency, Palm, or continuum claim.

4. Object-specific hostile checks

- Recompute both critical values and the orientation of the positive gap.
- Verify the conversion from λ to r without importing a physical-field normalization.
- Try the sign-reversed and rescaled cubic mutations and confirm they fall outside the stated scope.
- Attempt at least one additional mutation that changes an orientation, sign, normalization, law, endpoint, or quantifier while leaving the text superficially similar.

5. Proof-obligation ledger

Obligation	Reviewer test	Initial state
O1 Object identity	Definitions, ordering, normalization, and law match the statement.	OPEN FOR REVIEW
O2 Hypotheses	Every hypothesis used is stated and no stronger premise is hidden.	OPEN FOR REVIEW
O3 Derivation	Each algebraic, analytic, topological, or probabilistic step is valid.	OPEN FOR REVIEW
O4 Quantifiers	Uniformity, existence, endpoint, and asymptotic quantifiers are correct.	OPEN FOR REVIEW
O5 Computation	Any executable or interval result is reproducible and non-vacuous.	OPEN FOR REVIEW
O6 Scope firewall	The conclusion is not silently enlarged beyond the exact object.	OPEN FOR REVIEW
O7 Contradictions	No known correction, counterexample, or retained caveat defeats the claim.	OPEN FOR REVIEW

Part II - Complete Controlling Source Text

The following is a paragraph-order-preserving text reproduction of the controlling Drive artifact. Typography is normalized for this packet; mathematical content is not intentionally rewritten. Use the manifest URL for native-file or byte-level checks.

GP-CLS-PROP-006-v1.0 — PROPOSED TERMINAL CLOSURE OF THE CANONICAL A_2 FOLD-GAP IDENTITY

Artifact ID: GP-CLS-PROP-006-v1.0

Author: OpenAI GPT-5.6 Thinking

Created: 2026-07-21

Class: CLS-PROP — complete closure package awaiting human terminal approval

Authority: none until Dylan M. Roy accepts the terminal closure

Canonical impact: NONE

Proposed terminal label: CLOSED — EXACT CANONICAL NORMAL-FORM IDENTITY

1. Exact object proposed for closure

For the one-dimensional canonical A_2 fold family

$$f_{\lambda}(x) = -x^3/3 + \lambda x, \quad \lambda > 0,$$

the two critical points are $x_- = -\sqrt{\lambda}$ and $x_+ = +\sqrt{\lambda}$. Let $r = x_+ - x_- = 2\sqrt{\lambda}$ be their separation and let $\ell = f_{\lambda}(x_+) - f_{\lambda}(x_-)$ be the positive critical-value gap. Then

$$\ell = r^3/6.$$

The proposed closure applies only to this canonical normal-form statement and its exact algebraic consequences.

2. Exact proof

Differentiating gives

$$f'_{\lambda}(x) = -x^2 + \lambda,$$

so the critical points are $x = \pm\sqrt{\lambda}$. Their values are

$$\begin{aligned} f_{\lambda}(\sqrt{\lambda}) &= -(\lambda^{3/2})/3 + \lambda^{3/2} = (2/3)\lambda^{3/2}, \\ f_{\lambda}(-\sqrt{\lambda}) &= -(\lambda^{3/2})/3 - \lambda^{3/2} = -(2/3)\lambda^{3/2}. \end{aligned}$$

Therefore

$$\ell = f_{\lambda}(\sqrt{\lambda}) - f_{\lambda}(-\sqrt{\lambda}) = (4/3)\lambda^{3/2}.$$

Since $r = 2\sqrt{\lambda}$, we have $\lambda = r^2/4$ and hence

$$\begin{aligned} \ell &= (4/3)(r^2/4)^{3/2} \\ &= (4/3)(r^3/8) \\ &= r^3/6. \end{aligned}$$

No limiting argument, statistical fit, numerical approximation, or literature import is used.

3. Evidence and independent review

Primary derivation:

- CL-DER-012-v1.0, §1, Drive ID 1-V6YBESa8UV1Jy4Dh1IQrsp4Q80JilYqh4p2FrtEYqA.

Independent review:

- GP-AUD-008-v1.0 independently verified the canonical A_2 algebra and distinguished it from broader exponent and physical-field claims.
- CL-ERR-013 explicitly corrected the unstable separation-exponent claims while leaving the fold algebra intact.
- CL-AUD-021 reports an additional hostile verification of the corresponding CW derivation.

This package also independently re-derived the identity from first principles above.

4. Scope boundary — binding

This closure does NOT assert that $1/6$ is a universal coefficient for an arbitrary smooth physical Gaussian field in unnormalized coordinates.

For a general field, local coordinate changes and the cubic-jet normalization alter the physical relation between the measured separation and critical-value gap. The governing program therefore retains the mark

$$\kappa = 6\ell/r^3$$

when discussing the physical-field object. The exact closure is:

$$\text{canonical } A_2 \text{ coordinate} + \text{canonical cubic normalization} \Rightarrow \ell = r^3/6.$$

It is not:

$$\text{every physical maximum-saddle pair} \Rightarrow \ell = r^3/6.$$

5. Dependencies affected

This closure would retire repeated review of:

- whether the four-line A_2 algebra is exact;
- whether the coefficient $1/6$ is fitted or tunable inside the canonical normal form;
- whether the Jacobian calculation may use $d\ell/dr = r^2/2$ under that canonical convention;
- the narrow identity portion of CL-DER-012 and its repeated re-verifications.

6. Dependencies unaffected

This closure does not establish or alter:

- Theorem B or the $-1/3$ lifetime-density theorem;
- the candidate-pair separation measure;
- adjacency positivity, elder-rule selection, q-objects, or Palm-law transfers;
- the physical-field κ distribution or its tails;
- numerical continuum validation;

- P0.1, P0.2, P1.1, or P1.2;
- any canonical theorem or active machine state.

7. Terminal disposition requested

Requested terminal label:

CLOSED — EXACT CANONICAL NORMAL-FORM IDENTITY

Exact terminal statement:

“For $f_\lambda(x) = -x^3/3 + \lambda x$ with $\lambda > 0$, the critical-point separation $r = 2\sqrt{\lambda}$ and positive critical-value gap ℓ satisfy exactly $\ell = r^3/6$. This statement is confined to the canonical A_2 coordinate and normalization. Physical-field claims must retain the κ mark or an equivalent normalization factor.”

8. Register and ledger updates after human approval

If Dylan approves this terminal statement:

- move EC-005 from the Easy Closure Queue to the Closure Log;
- register the exact statement and scope boundary;
- mark repeated algebra-verification requests terminal;
- preserve all source derivations, errata, and audits;
- do not promote any broader exponent or theorem claim.

9. Human approval requirement

Human approval: REQUIRED.

Reason: this is a mathematical terminal closure intended to remove the item permanently from active review. The proof is complete, but OP-PROT-004 reserves final authority over what counts as definitively closed to Dylan M. Roy.

10. Correction path

A future correction may reopen only the wording or coordinate convention if evidence shows that the declared canonical family, separation orientation, or gap sign was misstated. Any correction must be additive. It must not reopen unrelated numerical or continuum claims already excluded by the scope boundary.

Part III - Terminal-State Reconciliation

1. Closure decision

Exact internal disposition	CLOSED - EXACT CANONICAL NORMAL-FORM IDENTITY
Closure date	2026-07-21
Evidence basis	Exact symbolic derivation; independent recomputation; hostile scope audit distinguishing the canonical coefficient from the physical-field kappa mark.
Independent review already recorded	AO48 and Anthropic/CL independently recomputed the identity; terminal condition was discharged before closure.

The prior closure record is evidence about the research program's internal state, not a substitute for your review. A new material objection should be reported even if the internal label is CLOSED.

2. Correction and reopening rule

Reopen or amend this object if you identify a counterexample, an object-definition mismatch, a sign/factor/quantifier defect, an unreproducible decisive computation, a hidden premise, or a scope claim not supported by the proof. Corrections must be additive and must preserve the historical source and review record.

3. Limits

- No universal physical-field coefficient; no lifetime theorem; no adjacency, Palm, or continuum claim.
- A correct lemma may still be unusable downstream if another dependency uses a different field, law, coordinate convention, branch mark, normalization, or parameter domain.
- A prior independent review does not reduce your duty to reconstruct the argument; duplicate lineages must not be counted as additional independence.

Part IV - Reproduction Instructions

- Record your name, institution or affiliation, expertise, date, and any prior exposure to this research program.
- Reconstruct the exact statement from Part I without relying on the historical verdict.
- Audit Part II line by line. For computational objects, obtain the raw carrier from the manifest, verify hashes first, inspect source safety, then execute in a clean environment.
- Complete the obligation ledger and hostile checks. Preserve exact counterexamples, code, logs, and environment details.
- Return the verdict form below plus a finding list with severity: critical, material, clarification, editorial, or no issue.

Part V - Reviewer Verdict Form

Reviewer	_____
Affiliation / expertise	_____
Prior exposure	None / limited / substantial - explain: _____
Independent reconstruction	Complete / partial / not possible
Verdict	APPROVE / APPROVE WITH CLARIFICATIONS / AMEND REQUIRED / REJECT / CANNOT VERIFY
Exact scope accepted	_____
Critical or material findings	_____ _____
Reproduction artifacts	Code/log/hash/location: _____

Signature and date

Required narrative: summarize your independent derivation; list every disagreement; state whether each issue affects the theorem statement, only the proof, only reviewability/reproducibility, or only presentation. State explicitly which excluded claims you did not review.

Part VI - Manifest and Provenance

Controlling source: https://docs.google.com/document/d/10c4ryKQHDk6MROJfh8zeFPN5gjq576pKN2kUTp2_mDA/edit

Closure / terminal record: https://docs.google.com/document/d/1Zm65j9BZe_z3P_8qP7rfZukRU-IWUE8h8bDqXdDsIQ/edit

Local source snapshot used to construct this packet: GP-CLS-PROP-006-v1.0 — Proposed terminal closure of the canonical A_2 fold-gap identity

Local source snapshot SHA-256 (DOCX export, transport checksum only):
bcda56c3749ca56c7c15133e0c7e85c3ce6417a2e92ddff63f1e331dd718338b

Authority note: the transport checksum above identifies the locally downloaded Office export used for packet construction. It is not substituted for any marker-delimited frozen-body hash declared inside the source artifact.

Supplemental Drive access may be granted for raw capsules, revision history, comments, exact native exports, and dependency artifacts. The packet is designed so that those materials strengthen reproducibility rather than supply missing mathematical context.